WEIGHTED GENERALIZED MEAN POOLING FOR DEEP IMAGE RETRIEVAL

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ABSTRACT

Spatial pooling over convolutional activations (e.g., max pooling or sum pooling) has been shown to be successful in learning deep representations for image retrieval. However, most pooling techniques assume that every activation is equally important, and as a result they suffer from the presence of uninformative image regions that play a negative role as regards matching or lead to the confusion of particular visual instances. To address this issue, we propose a trainable building block that steers pooling to local information important to the task at hand. The method formulates pooling as a weighted generalized mean (wGeM), in which weights are learned on activations, reflecting the discriminative power of each activation in image matching. Embedding wGeM in a deep network improves image representation and boosts retrieval performance on standard benchmarks. wGeM does not require any bounding box annotations, but instead learns the latent probabilities of activations from scratch. It even goes beyond objectness, and learns to look at important visual details rather than the whole region of the object of interest.

Index Terms — Deep convolutional networks, image retrieval, learning to rank, pooling, weighted generalized mean

1. INTRODUCTION

Fine-tuning a deep network to learn deep image representations [1–6] has been shown to result in a significant improvement in image retrieval over off-the-shelf descriptors [7–11]. Even more crucial is to optimize the network directly with the similarity to be used in the retrieval task by combining 1) a good pre-trained classification network, 2) a good pooling method, and 3) a ranking loss [2–4]. Previously, a number of pooling methods were used, ranging from fully connected layers [1,7] to global pooling methods including sum pooling [8], max pooling [3, 9, 11], generalized mean (GeM) pooling [6], regional pooling [5,9], and aggregation techniques [2,12,13]. Most of these methods perform uniform pooling over the activations of convolutional layers, in which each activation contributes equivalently to the construction of a global representation.

However, spatially uniform pooling suffers from the presence of activations that play a negative role as regards matching or lead to the confusion of particular visual instances. To handle this problem, Gordo et al. [5] proposed a region proposal network (RPN) trained to localize objects of interest in images. RPN learns to predict region proposals at an object level where not all the information has to be important to the final retrieval task. It also requires bounding box annotations for training and a non-trivial amount of preprocessing effort. Kalantidis et al. [10] applied spatial weighting before sum pooling to boost locations at which a larger number of strong activations co-occurred and to down-weight less active locations. Nevertheless, the weighting layer in their work is non-trainable, and the activation strength may not correlate with the discriminative power of locations.

In this paper, we propose exploiting a spatial weighting mechanism for pooling over the activations of the last convolutional layer. This approach predicts a weight that describes how discriminating each activation at each location is as regards image matching and retrieval. Our idea is inspired by visual attention, which has been successfully employed in image captioning [14], machine translation [15], and object recognition [16]. We introduce this mechanism into the image retrieval task by extending it and incorporating it with pooling and learning to rank, leading to the end-to-end learning of deep representations based on a novel weighted generalized mean (wGeM) pooling method. Sum pooling [8], max pooling [9,11], and GeM pooling [6] are all special cases of wGeM. Our method also generalizes Kalantidis's method [10], while making the weighting mechanism trainable. Unlike RPN [5], which predicts rigid bounding boxes for objects, wGeM learns the latent probabilities of activations from scratch. Requiring no bounding box annotations for training, our method dispenses with the need for costly preprocessing or any assumption regarding the training dataset. We demonstrate that incorporating wGeM into a deep network improves image representation and boosts retrieval performance on standard benchmarks.

The rest of the paper is organized as follows. Our network architecture and the proposed wGeM block are described in Section 2. We conduct evaluations of wGeM on multiple datasets, and compare the results with those for the state of the art approach in Section 3. Conclusions and future directions are given in Section 4.

2. PROPOSED METHOD

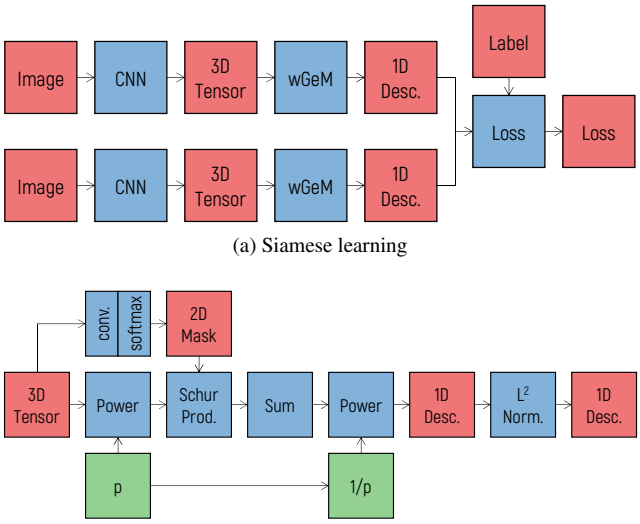
2.1. Network Architecture and Learning

The proposed wGeM can be applied to any convolutional neural network (CNN), e.g., AlexNet [17], VGG [18], and ResNet [19]. Such a CNN trained for image classification provides a good initialization when performing fine-tuning for retrieval [4–6]. Given an image $\mathbf{I}$, the output of the CNN, whose fully connected layers are discarded, is a 3D tensor $\mathbf{X} \in \mathbb{R}^{H \times W \times K}$ with K being the number of channels. We now add a wGeM block that takes $\mathbf{X}$ as an input and produces an l^2-normalized descriptor $\bar{\mathbf{y}} \in \mathbb{R}^K$ as an output. The details of the wGeM are provided in Section 2.2.

In this paper, we follow Radenovic et al. [6] as regards the learning procedure. A Siamese network is adopted as shown in Fig. 1a, in which the two branches share the same parameters. The training input consists of image pairs $(\mathbf{I}_i, \mathbf{I}_j)$ and labels $z_{i,j} \in \{0, 1\}$ declaring whether a pair is non-matching or matching, respectively. Let $\bar{\mathbf{y}}_i$ be the descriptor of image $\mathbf{I}_i$ output from the wGeM. We employ the contrastive loss [3, 20], which is defined as

$$\mathcal{L}_{i,j} = \begin{cases} \frac{1}{2}\|\bar{\mathbf{y}}_i - \bar{\mathbf{y}}_j\|_2^2 & \text{if } z_{i,j} = 1 \\ \frac{1}{2}[\max(0, \tau - \|\bar{\mathbf{y}}_i - \bar{\mathbf{y}}_j\|_2)]^2 & \text{otherwise,} \end{cases} \quad (1)$$

where τ is a hyper-parameter defining the margin of the hinge loss such that non-matching pairs pay a linear penalty only when the Eu-



(a) Siamese learning

(b) Our wGeM building block

Fig. 1. Architecture of our network and wGeM pooling. Blocks and layers are shown in blue. Variables and parameters are shown in red and green, respectively.

clidean distances are smaller than τ. The network is trained using a large number of training pairs that are created automatically [3, 21, 22]. During testing, the fine-tuned network produces a descriptor $\bar{\mathbf{y}}$ for the query and for each document in the corpus. Retrieval proceeds by computing the similarity between the query and each document descriptor using the Euclidean distance.

2.2. Weighted Generalized Mean Pooling

It should be recalled that wGeM takes the tensor $\mathbf{X}$ as an input and produces a 1D descriptor $\bar{\mathbf{y}}$. Let $\mathbf{x}_k \in \mathbb{R}^{H \times W}$ be the activation map for channel $k \in \{1, \ldots, K\}$. Let $N = H \times W$. We reshape $\mathbf{x}_k$ such that $\mathbf{x}_k = [x_{1,k} \ldots x_{N,k}]^\top$ for clarity. It is also assumed that the last layer of the CNN is a ReLU such that all $\mathbf{x}_k \in \mathbf{X}$ are non-negative.

Given a sequence of positive weights $\mathbf{\Omega} = [\omega_1 \ldots \omega_N]^\top$ with $\|\mathbf{\Omega}\|_1 = 1$, we exploit the weighted generalized mean and propose to construct a 1D descriptor $\mathbf{y} = [y_1 \ldots y_K]^\top$ in the form of

$$y_k = \left(\sum_{i=1}^{N} \omega_i x_{i,k}^p \right)^{1/p}, \tag{2}$$

where p is the exponent. This is the core of our wGeM. Each weight $\omega_i \in \mathbf{\Omega}$ describes how contributive the activations $[x_{i,1} \ldots x_{i,K}]^\top$ at location i are to distinguishing image $\mathbf{I}$ from others. $\mathbf{\Omega}$ can thus be understood as a soft 2D mask. The wGeM is related to visual attention mechanisms [14–16] but it is distinguished from them by the presence of the exponent p. The descriptor $\mathbf{y}$ is later l^2-normalized, leading to the final descriptor $\bar{\mathbf{y}}$. The similarity between two images thus becomes a cosine similarity and can be computed with the inner product $\bar{\mathbf{y}}_i^\top \bar{\mathbf{y}}_j$. For widely-used CNNs [18, 19], K equals 512 or 2,048, making $\bar{\mathbf{y}}$ a compact image representation.

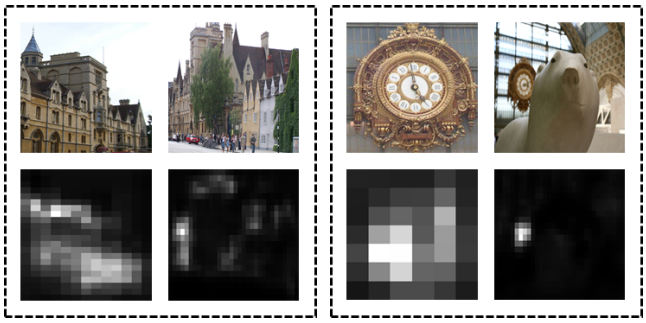
Following Radenovic et al. [6], we learn the exponent p along with the fine-tuning of the CNN. We consider a function $\Phi : \mathbf{X} \mapsto \mathbf{\Omega}$ that parameterizes $\mathbf{\Omega}$ by transforming $\mathbf{X}$ into a flexible decision regarding whether or not to encode the activations at location i into $\mathbf{y}$. As with p, Φ can be learned with fine tuning. Although it can be based on any CNN block, we chose a single 3×3 convolution followed by a spatial softmax function for this purpose, as shown in Fig. 1b. This single convolution alleviates the increase in back-propagation complexity. We adopt a softmax function instead of a sigmoid or ReLU function so that $\mathbf{\Omega}$ satisfies $\|\mathbf{\Omega}\|_1 = 1$. The derivatives of Eq. 2 with respect to the variables x and ω and the parameter p are given by Eqs. 3, 4, and 5, respectively. On the right side of Eq. 3, the first and the second terms correspond to back-propagation through the upper and middle paths in Fig. 1b, respectively.

$$\frac{\partial y_k}{\partial x_{i,k}} = \sum_{j=1}^{N} \frac{\partial y_k}{\partial \omega_j} \frac{\partial \omega_j}{\partial x_{i,k}} + \omega_i \left(\frac{x_{i,k}}{y_k} \right)^{p-1} \tag{3}$$

$$\frac{\partial y_k}{\partial \omega_i} = \frac{x_{i,k}}{p} \left(\frac{x_{i,k}}{y_k} \right)^{p-1} \tag{4}$$

$$\frac{\partial y_k}{\partial p} = \frac{y_k}{p} \left(\frac{\sum_i \omega_i x_{i,k}^p \log x_{i,k}}{y_k^p} - \log y_k \right) \tag{5}$$

Our spatial weighting mechanism is expected to reflect the discriminative power of the activations at each location. Two matching examples are shown in Fig. 2. In the right image in Fig. 2a, the wGeM well localized the object of interest, i.e., Balliol College, and successfully disregarded occluding objects, e.g., trees and houses. Figure 2b shows a more challenging example, in which the images differ from each other with large viewpoint and scale changes.



(a) (b)

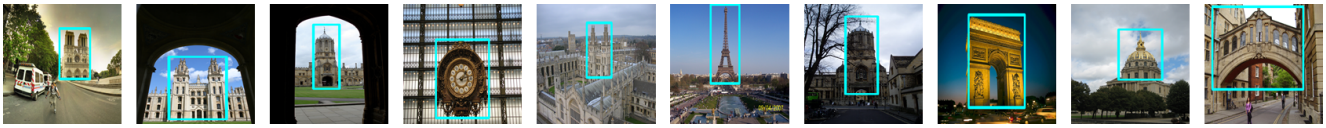
Fig. 2. Matching images and their 2D masks $\mathbf{\Omega}$ produced by wGeM fine-tuned with ResNet101 [19] on independent dataset.

Table 1. Comparisons of pooling methods for deep representation. Our method is marked with $\star$. We use wGeM to denote both our method and the weighted generalized mean.

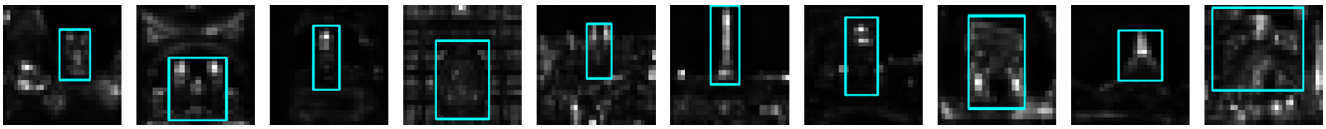
	ω	p	**Pooling**
SPoC [8]	$1/N$	1	Sum
CroW [10]	$\sum_k x_{i,k}$	1	wSum
MAC [3, 9, 11]	–	∞	Max
GeM [6]	$1/N$	Parameter	GeM
$\star$ wGeM	$\Phi(\mathbf{X})$	Parameter	wGeM

2.3. Properties of wGeM

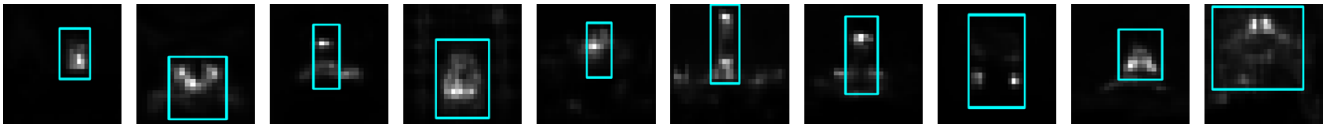
As shown in Table 1, the wGeM generalizes previous pooling methods, e.g., sum pooling [8], weighted sum pooling [10], max pooling [3, 9, 11], and GeM pooling [6]. Unlike GeM pooling [6], which is based on uniform pooling, the wGeM ignores the redundant part



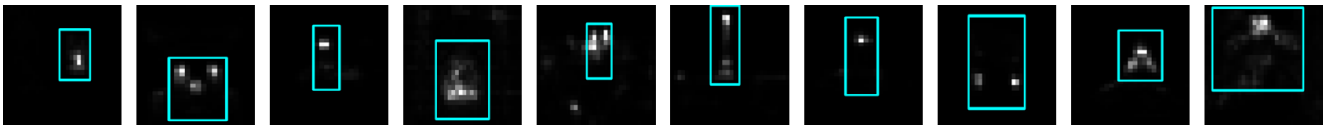
(a) Images from Oxford5K [23].



(b) Location-wise l^1-norm of 3D tensor $\mathbf{X}$ raised to power of p, i.e., $\sum_k x_{i,k}^p$



(c) $\mathbf{\Omega}$ predicted by wGeM



(d) Location-wise l^1-norm of 3D tensor $\mathbf{X}$ raised to power of p and then weighted by $\mathbf{\Omega}$, i.e., $\sum_k \omega_i x_{i,k}^p$

Fig. 3. Activation strengths before and after spatial weighting. Each column corresponds to an image in (a). The objects of interest predefined in Oxford5K [23] are outlined in blue. The CNN initialized by ResNet101 [19] was fine-tuned on an independent landmark dataset [3, 21, 22].

of an image while encoding it into a descriptor. The wGeM can also be applied to regional pooling [5, 9], in which the network produces multiple $\mathbf{\Omega}$, each being associated with a single region. Each $\mathbf{\Omega}$ can either be directly obtained from the activations in each region or be sampled from the global $\mathbf{\Omega}$ obtained from $\mathbf{X}$.

One important property of the wGeM relates to its behavior when $p \to \infty$. In this case, Eq. 2 reduces to max pooling regardless of ω. During back-propagation, the derivative of the loss $\mathcal{L}$ with respect to ω is

$$\frac{\partial \mathcal{L}}{\partial \omega_i} = \frac{1}{p} \sum_{k=1}^{K} \frac{\partial \mathcal{L}}{\partial y_k} \left(\frac{x_{i,k}}{y_k} \right)^{p-1} x_{i,k}, \tag{6}$$

where $\left(\frac{x_{i,k}}{y_k} \right)^{p-1} \in \{0, 1\}$ and equals one only if $y_k = x_{i,k}$. Eq. 6 thus approaches zero as $\frac{1}{p} \to 0$. That is, all $\omega_i \in \mathbf{\Omega}$ become non-functional during both forward and backward propagations. A similar issue exists with the derivative of $\mathcal{L}$ with respect to p. In practice, the learned p never reached a large value in our experiments, which is in accordance with the results of GeM pooling [6].

3. EXPERIMENTS

3.1. Experimental Setup

ResNet101 [19] was used for the initialization of the convolutional layers of our network. As in Radenovic's studies [3, 6], the network was trained using a dataset of image tuples derived from more than 163K images [3, 21, 22]. Training was performed for 30 epochs and the best network was selected on the basis of its MAP on a validation set. For each epoch, around 6K and 1.7K tuples were selected for training and validation, respectively. Each tuple contains one query, one positive, and five negative images. Adam [24] was used with an initial learning rate of $l_0 = 10^{-6}$, an exponential decay $l_0 e^{-0.1i}$ over epoch i, and a batch size of five tuples. The margin τ in Eq. 1 was set at 0.85. As for the 3×3 convolutional layer that produces the 2D mask, both the weights and the bias of the filter were initialized with zeros such that $\forall \omega_i \in \mathbf{\Omega}$, $\omega_i = \frac{1}{N}$ in the first iteration.

Table 2. MAPs of wGeMs obtained with different initializations.

	Initial p	Oxford5K [23]	Paris6K [25]
GeM [6]	3	87.8	92.7
wGeM	2	88.6	92.2
	3	88.9	92.3
	4	88.8	92.5
	5	88.4	92.6

For evaluation, we used three benchmarks that are widely used in instance-level image retrieval: Oxford5K [23], Oxford105K [23], and Paris6K [25]. The mean average precision (MAP) was used as a performance measure for all the datasets. The supervised whitening and the multi-scale representation schemes, proposed by Radenovic et al. [3, 6], were both adopted for improved retrieval.

3.2. Results

We first analyzed the wGeM pooling performance by comparing it with the GeM [6]. In Eq. 2, the exponent p can be configured in various ways, e.g., 1) using a single p shared by all channels, 2) using channel-dependent p_k with $k \in \{1, \ldots, K\}$, and 3) regarding p as a hyper-parameter. With uniform pooling, Radenovic et al. [6] showed that learning a shared p performed better than the other solutions.

Table 3. MAP comparison with state-of-the-art image retrieval based on deep networks. Our method is marked with $\star$, in which p is initialized by 4. The updated state-of-the-art results are highlighted in bold.

	CNN	Fine Tuned	QE	Oxford5K [23]	Oxford105K [23]	Paris6K [25]
SPoC [6, 8]	VGG	✗	✗	56.4	47.8	72.3
CroW [10]	VGG	✗	✗	70.8	65.3	79.7
BoW-CNN [12]	VGG	✗	✗	73.9	59.3	82.0
NetVLAD [2]	VGG	✓	✗	71.6	–	79.7
MAC [3]	VGG	✓	✗	80.0	75.1	82.9
Fisher Vector [13]	VGG	✓	✗	81.5	76.6	82.4
R-MAC [5]	ResNet	✓	✗	86.1	82.8	**94.5**
GeM [6]	ResNet	✓	✗	87.8	84.6	92.7
$\star$ wGeM	ResNet	✓	✗	**88.8**	**85.6**	92.5
CroW+QE [10]	VGG	✗	✓	74.9	70.6	84.8
BoW-CNN+QE [12]	VGG	✗	✓	78.8	65.1	84.8
MAC+QE [3]	VGG	✓	✓	85.4	82.3	87.0
R-MAC+QE [5]	ResNet	✓	✓	90.6	89.4	**96.0**
GeM+αQE [6]	ResNet	✓	✓	91.0	89.5	95.5
$\star$ wGeM+QE	ResNet	✓	✓	**91.7**	**89.7**	**96.0**

Fig. 4. Example of spatial weighting failure with wGeM. In this pair of matching images, the wGeM looked towards contexts outside the object of interest.

We adopted the same strategy and evaluated the effect of different initializations of p during fine tuning. The results are shown in Table 2. The wGeM consistently outperformed the conventional GeM for Oxford5K, but the MAP was comparable to or slightly poorer for Paris6K depending on the initial value of p. Looking at the training data, we found that the training set is composed of more historical landmarks, which is the same as Oxford5K, while Paris6K contains more contemporary buildings, offering complexity and a variety of different styles. Therefore, the wGeM may focus on activations that are discriminating for describing images in Oxford5K but less helpful for Paris6K. Later we will show that this issue can be alleviated with the widely adopted query expansion approach.

Figure 3 shows how the activations are affected by the weighting mechanism. In previous studies [10, 26], it has been shown that the activations themselves already have a high spatial correlation with objects on the image level. This is partially reflected in Fig. 3b. However, although the CNN has been fine-tuned on landmark images, it still fires to background clutter, uncommon objects, e.g., the car in the leftmost example, and unhelpful finely-textured patterns, e.g., the background in the fourth example. In Figs. 3c and 3d, we can see how the 2D masks predicted by the wGeM better localize the objects of interest and improve the activations.

We compared our results extensively with state-of-the-art image retrieval based on deep representations. Table 3 summarizes the wGeM results together with previously reported MAPs. We have the state-of-the-art MAPs on Oxford5K and Oxford105K, while our method is outperformed by the work of Gordo et al. [5] on Paris6K. Unlike Gordo's study, which involves manually acquired landmark labels, our method does not require any manual labeling or costly cleansing of training data. Moreover, we combined our method with a query expansion (QE) method [27], which has been widely used for deep image retrieval [3, 5, 8–10]. Here, an initial search is issued with the query, and the top n_{QE} results are retrieved. The deep descriptors of the original query and the top results are then sum-aggregated and renormalized. A second-round search is thus issued with the renormalized descriptor, producing the final ranking list of retrieved images. We set $n_{\mathrm{QE}} = 10$ for Oxford5K and Oxford105K and $n_{\mathrm{QE}} = 50$ for Paris6K. Once combined with QE, the proposed method further boosts performance, and outperforms or is on a par with the state of the art on all datasets.

Figure 4 shows a case where our wGeM for spatial weighting fails. Here, the object of interest, the Louvre Pyramid, was largely ignored by the wGeM, and the adjacent building attracted most of the attention. A potential reason may be that regular patterns are commonly used in building design and are potentially less discriminating when matching the landmarks in the training set. Therefore, the wGeM was trained to fire less for such regions. This issue may be alleviated by introducing more object classes, for which regular textures play a much more prominent role in discrimination, in the training set.

4. CONCLUSIONS

We addressed image retrieval with the learning of deep representations. We proposed a novel pooling method based on a spatial weighting mechanism, which sorts out discriminating information from the activation tensor output from any fully convolutional CNN. Compared with previous methods based on uniform pooling [3, 6, 8–11], our proposed wGem pooling is more robust as regards unhelpful visual details that play a negative role as regards matching or lead to the confusion of particular visual instances. Our method achieved accuracies of 91.7% and 89.7% on Oxford5K and Oxford105K, respectively, and updated the state-of-the-art approach based on deep networks. In the future, we shall investigate how different wGeM block structures affect the learning of deep representations.

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